

A SQUARE COUNTEREXAMPLE TO CRAWFORD-NUMBER LOCALIZATION NEAR CONVEX GENERATORS

AGENT LANE 04

STATUS

This packet gives a finite-dimensional counterexample to Question 3.2 in Choi–Lee, *On the Crawford number attaining operators* [1]. The question asks whether, whenever $S \subset B_X$ and $B_X = \overline{\text{co}}(S)$, the Crawford number $c(T)$ can be computed by restricting the infimum to states (x, x^*) with $\text{dist}(x, S) < \varepsilon$. The answer is no, even for a two-dimensional real Banach space and a finite uniformly strongly exposed generating set S .

SOURCE QUESTION

The source asks:

Let X be a Banach space and $S \subset B_X$ be a subset such that $B_X = \overline{\text{co}}(S)$. Given $\varepsilon > 0$ and $T \in \mathcal{L}(X)$, does

$$c(T) = \inf\{|x^*(Tx)| : (x, x^*) \in \Pi(X), \text{dist}(x, S) < \varepsilon\}$$

hold?

ANSWER IN $\mathcal{L}(X)$.

3.2. Property α . Our next approach is to determine the denseness of Crawford number attaining operators when X has so-called property α . A Banach space X is said to have *property α* if there exist $\{x_\alpha\}_{\alpha \in I} \subseteq S_X$ and $\{x_\alpha^*\}_{\alpha \in I} \subseteq S_{X^*}$ with a constant $0 \leq \rho < 1$ satisfying that

- (i) $x_\alpha^*(x_\alpha) = 1$ for all $\alpha \in I$,
- (ii) $|x_\alpha^*(x_\beta)| \leq \rho < 1$ with $\alpha, \beta \in I$ with $\alpha \neq \beta$,
- (iii) $B_X = \overline{\text{co}}\{x_\alpha\}_{\alpha \in I}$, where $\overline{\text{co}}$ denotes the closed convex hull of a set.

One can easily show that $B_X = \overline{\text{co}}(S)$ for some uniformly strongly exposed set S (see [18] for the definition of a uniformly strongly exposed set) if X has property α . Thus by following the repertoire of [1], we may find a way to show the denseness of Crawford number attaining operators when X has property α . The main difference here, compared to ones in [1], would be the following one.

Question 3.2. Let X be a Banach space and $S \subseteq B_X$ be a subset such that $B_X = \overline{\text{co}}(S)$. Given $\varepsilon > 0$ and $T \in \mathcal{L}(X)$, does

$$c(T) = \inf\{|x^*(Tx)| : (x, x^*) \in \Pi(X), \text{dist}(x, S) < \varepsilon\}$$

hold?

DEFINITIONS

For a Banach space X , the state space is

$$\Pi(X) = \{(x, x^*) \in S_X \times S_{X^*} : x^*(x) = 1\}.$$

For $T \in \mathcal{L}(X)$, the Crawford number is

$$c(T) = \inf\{|x^*(Tx)| : (x, x^*) \in \Pi(X)\}.$$

IDEA OF THE PROOF

In the square $X = \ell_\infty^2$, the four vertices generate the unit ball and are uniformly strongly exposed. However, the middle point of an edge can carry a state that sees the coordinate-swap operator as zero. Points close to the vertices have both coordinates close to signs ± 1 , and every norming state is supported on a coordinate where the point has modulus one. After the coordinate swap, that state sees the other coordinate, whose modulus is still close to one. Thus the global Crawford infimum is zero, but the infimum restricted near the generating set is bounded below by a positive number.

COUNTEREXAMPLE

Theorem 1. *Question 3.2 of [1] has a negative answer. More precisely, there are a real two-dimensional Banach space X , a finite uniformly strongly exposed set $S \subset B_X$ with $B_X = \text{co}(S)$, an operator $T \in \mathcal{L}(X)$, and $0 < \varepsilon < 1$ such that*

$$c(T) = 0$$

but

$$\inf\{|x^*(Tx)| : (x, x^*) \in \Pi(X), \text{dist}(x, S) < \varepsilon\} \geq 1 - \varepsilon > 0.$$

Proof. Let $X = (\mathbb{R}^2, \|\cdot\|_\infty)$ and let

$$S = \{(\sigma, \tau) : \sigma, \tau \in \{-1, 1\}\}$$

be the four vertices of the unit square. Then $B_X = \text{co}(S)$. The set S is uniformly strongly exposed: the vertex $v_{\sigma, \tau} = (\sigma, \tau)$ is exposed by

$$f_{\sigma, \tau}(a, b) = \frac{\sigma a + \tau b}{2}.$$

If $f_{\sigma, \tau}(a, b) > 1 - \eta$ for $(a, b) \in B_X$, then $\sigma a > 1 - 2\eta$ and $\tau b > 1 - 2\eta$, so $\|(a, b) - v_{\sigma, \tau}\|_\infty < 2\eta$. Since S is finite, this is a uniform strong exposure.

Define the coordinate-swap operator

$$T(a, b) = (b, a).$$

Let $x = (1, 0)$ and $x^* = e_1^*$, the first coordinate functional. Then $(x, x^*) \in \Pi(X)$ and

$$x^*(Tx) = e_1^*(0, 1) = 0.$$

Hence $c(T) = 0$.

Fix $0 < \varepsilon < 1$. Suppose $(x, x^*) \in \Pi(X)$ and $\text{dist}(x, S) < \varepsilon$. Write $x = (a, b)$. There are $\sigma, \tau \in \{-1, 1\}$ such that

$$|a - \sigma| < \varepsilon, \quad |b - \tau| < \varepsilon.$$

In particular $|a| > 1 - \varepsilon$ and $|b| > 1 - \varepsilon$.

Write $x^* = (\alpha, \beta) \in (\ell_\infty^2)^* = \ell_1^2$. Since $(x, x^*) \in \Pi(X)$,

$$|\alpha| + |\beta| = 1, \quad \alpha a + \beta b = 1.$$

The equality case in

$$\alpha a + \beta b \leq |\alpha| |a| + |\beta| |b| \leq |\alpha| + |\beta| = 1$$

implies that every coordinate on which x^* is supported is a norming coordinate of x , with matching sign. Thus, if $\alpha \neq 0$ then $|a| = 1$ and $\alpha a = |\alpha|$, while if $\beta \neq 0$ then $|b| = 1$ and $\beta b = |\beta|$.

Now

$$x^*(Tx) = \alpha b + \beta a.$$

If only the first coordinate is used by x^* , then $|x^*(Tx)| = |b| > 1 - \varepsilon$. If only the second is used, then $|x^*(Tx)| = |a| > 1 - \varepsilon$. If both are used, then $|a| = |b| = 1$ and the signs of a and b are σ and τ , respectively; in that case

$$x^*(Tx) = \sigma\tau(|\alpha| + |\beta|)$$

and so $|x^*(Tx)| = 1$.

Therefore every state with $\text{dist}(x, S) < \varepsilon$ satisfies $|x^*(Tx)| \geq 1 - \varepsilon$. The restricted infimum is positive, whereas $c(T) = 0$, proving the claimed failure. $\square$

LIMITATIONS

This counterexample answers Question 3.2 as stated, and even with S finite and uniformly strongly exposed. It does not settle the separate questions in the source about density of $\text{CNA}(X)$ for property α or property β spaces.

NOVELTY AND VERIFICATION NOTES

- Local run indexes were searched for 2201.10031, the source title, **Crawford number, property alpha**, and $\text{CNA}(X)$. No duplicate packet or exact attempt for this question was found.
- Bounded web search on 2026-06-29 for the source title and phrases such as “Crawford number property alpha” and “Crawford number closed convex hull” surfaced the source arXiv page but no later answer to this localization question.
- The example is finite dimensional and uses only the elementary description of states in ℓ_∞^2 .

HUMAN REVIEW RECOMMENDATION

Review as a likely-valid counterexample to Question 3.2. The key point to check is the elementary equality-case argument for states of ℓ_∞^2 . If accepted, the counterexample also shows that adding “ S uniformly strongly exposed” is not sufficient to rescue the localization formula.

REFERENCES

- [1] G. Choi and H. J. Lee, *On the Crawford number attaining operators*, arXiv:2201.10031, 2022. <https://arxiv.org/abs/2201.10031>